

SIMATS ENGINEERING
ASSESSMENT TOOL 2: LEVEL 2

COURSE NAME:	Programming in Java	COURSE CODE:	CSA0915- SLOT B
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ASSESSMENT TASKS:

Find the Mth maximum number and Nth minimum number in an array and then find the sum of it and difference of it.

Sample Input:

Array of elements = {14, 16, 87, 36, 25, 89, 34}

M = 1

N = 3

Sample Output:

1stMaximum Number = 89

3rdMinimum Number = 25

Sum = 114

Difference = 64

Testcases:

1. {16, 16, 16 16, 16}, M = 0, N = 1
2. {0, 0, 0, 0}, M = 1, N = 2
3. {-12, -78, -35, -42, -85}, M = 3 , N = 3

4. {15, 19, 34, 56, 12}, M = 6 , N = 3

5. {85, 45, 65, 75, 95}, M = 5 , N = 7



The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", the subtitle "JAVA PROGRAMMING", and a user profile for "S K DEEPA ARATHI" with ID "192571020RB". The main area is divided into "Your Code" and "Output".

```
1 import java.util.Scanner;
2
3 class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int[] arr = new int[n];
8
9         for (int i = 0; i < n; i++) {
10             arr[i] = sc.nextInt();
11         }
12
13         int max = arr[0];
14         int min = arr[0];
15
16         for (int i = 0; i < n; i++) {
17             if (arr[i] > max) {
18                 max = arr[i];
19             }
20             if (arr[i] < min) {
21                 min = arr[i];
22             }
23         }
24
25         int sum = min + max;
```

The output window displays the following results:

```
max element=90
min element=12
sum of elements=102
difference of elements=78
```

Write a program to count all the prime and composite numbers entered by the user.

Sample Input:

Enter the numbers

4

54

29

71

7

59

98

23

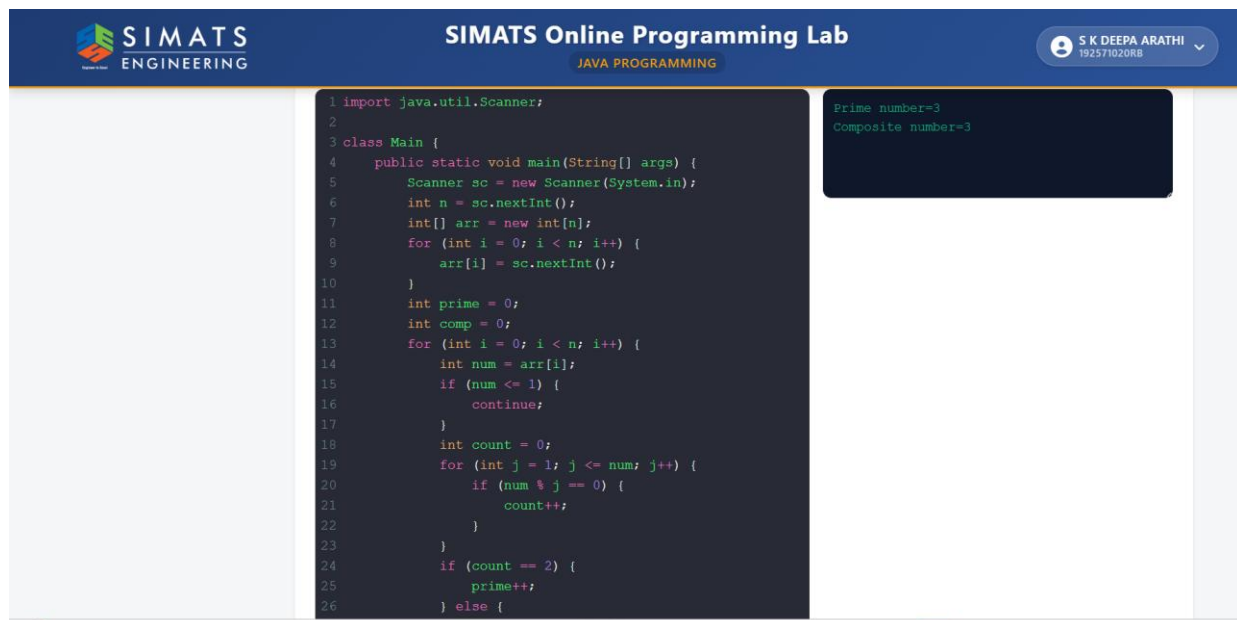
Sample Output:

Composite number:3

Prime number:5

Testcases:

1. 33, 41, 52, 61, 73, 90
2. TEN, FIFTY, SIXTY-ONE, SEVENTY-SEVEN, NINE
3. 45, 87, 09, 5.0, 2.3, 0.4
4. -54, -76, -97, -23, -33, -98
5. 45, 73, 00, 50, 67, 44



The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the text "SIMATS Online Programming Lab", and the user's name "S K DEEPA ARATHI" with a dropdown arrow. Below the header, the text "JAVA PROGRAMMING" is displayed. The main area contains a code editor with the following Java code:

```
1 import java.util.Scanner;
2
3 class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int[] arr = new int[n];
8         for (int i = 0; i < n; i++) {
9             arr[i] = sc.nextInt();
10        }
11        int prime = 0;
12        int comp = 0;
13        for (int i = 0; i < n; i++) {
14            int num = arr[i];
15            if (num <= 1) {
16                continue;
17            }
18            int count = 0;
19            for (int j = 1; j <= num; j++) {
20                if (num % j == 0) {
21                    count++;
22                }
23            }
24            if (count == 2) {
25                prime++;
26            } else {
```

On the right side of the code editor, there is a small output window displaying the results of the program:

```
Prime number=3
Composite number=3
```

Write a program to print the total amount available in the ATM machine with the conditions applied.

Total denominations are 2000, 500, 200, 100, get the denomination priority from the user and the total number of notes from the user to display the total available balance to the user

Sample Input:

Enter the 1st Denomination: 500

Enter the 1st Denomination number of notes: 4

Enter the 2nd Denomination: 100

Enter the 2nd Denomination number of notes: 20

Enter the 3rd Denomination: 200

Enter the 3rd Denomination number of notes: 32

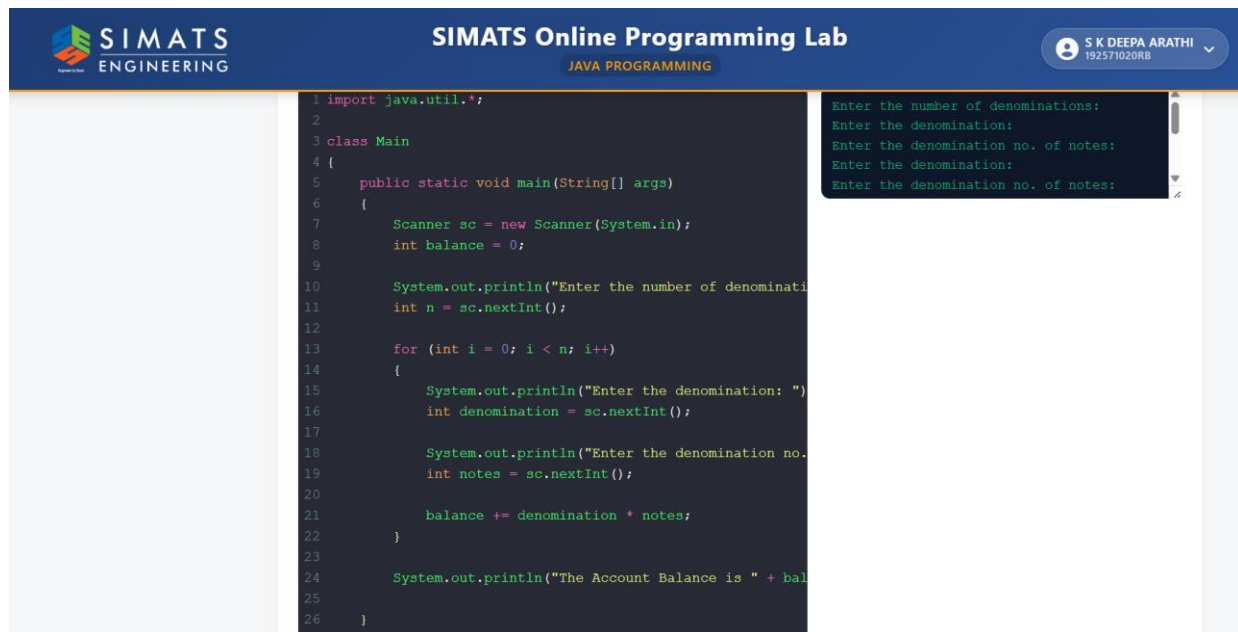
Enter the 4th Denomination: 2000

Enter the 4th Denomination number of notes: 1

Sample Output:

Total Available Balance in ATM: 12400

Testcases:



The screenshot displays the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the text "SIMATS Online Programming Lab", and the user's name "S K DEEPA ARATHI" with a dropdown arrow. Below the header, the "JAVA PROGRAMMING" tab is selected. The main area is divided into two panels. The left panel shows a Java program for calculating the total available balance in an ATM based on the number of denominations and the number of notes for each denomination. The right panel shows the sample output of the program, which matches the input and output provided in the text above.

```
1 import java.util.*;
2
3 class Main
4 {
5     public static void main(String[] args)
6     {
7         Scanner sc = new Scanner(System.in);
8         int balance = 0;
9
10        System.out.println("Enter the number of denominations:");
11        int n = sc.nextInt();
12
13        for (int i = 0; i < n; i++)
14        {
15            System.out.println("Enter the denomination: ");
16            int denomination = sc.nextInt();
17
18            System.out.println("Enter the denomination no. of notes:");
19            int notes = sc.nextInt();
20
21            balance += denomination * notes;
22        }
23
24        System.out.println("The Account Balance is " + balance);
25    }
26 }
```

Enter the number of denominations:
Enter the denomination:
Enter the denomination no. of notes:
Enter the denomination:
Enter the denomination no. of notes:

Write a program using choice to check

Case 1: Given string is palindrome or not

Case 2: Given number is palindrome or not

Sample Input:

Case = 1

String = MADAM

Sample Output:

Palindrome

Testcases:

1. MONEY

2. 5678765

3. MALAY12321ALAM

4. MALAYALAM

5. 1234.4321

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JAVA PROGRAMMING

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